

Juan F. Fung

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Employment

National Institute of Standards and Technology

Gaithersburg, MD

Economist - Applied AI Research Group, Information Technology Laboratory June 2026 - present

- Lead research on AI measurement and evaluation for frontier AI capabilities, including program evaluation and measurement frameworks for impacts of AI adoption.
- Build AI-powered research tools, including synthetic-data generation agents and survey instrument design and validation.
- Lead multi-university research collaborations applying Large Language Models (LLMs) to topic modeling and analysis of large text corpora.

National Institute of Standards and Technology

Gaithersburg, MD

Economist - Applied Economics Office, Engineering Laboratory September 2016 - June 2026

- Led interdisciplinary research on the economics of disaster resilience, developing predictive models for seismic retrofit costs, benefit-cost frameworks for functional recovery design, and community resilience measurement tools, producing 19 peer-reviewed publications and multiple technical reports adopted by practitioners.
- Served as contributor on the EL AI Goal Tiger Team, authoring the vision statement and AI goal for the EL Goal Charter to support AI research in EL; received a Special Act Award. Secured competitive internal funding and designed survey instruments and analytical frameworks for evaluating organizational policy changes.
- FY22-FY23: Served as Analyst in NIST Director's Office, developing interactive data visualization dashboards, facilitating cross-functional data governance initiatives, and representing NIST in Department and interagency working groups.

Education

University of Illinois

Urbana-Champaign, IL

PhD Economics, MS Statistics

August 2016

- Dissertation: "Three Essays in Applied Market Design"

University of Missouri-St. Louis

St. Louis, MO

MS Economics

May 2008

New York University

New York, NY

BA Economics

May 2003

Publications (peer-reviewed)

1. Fung, J. F., Cook, D. T., Elsibaie, S., Zhang, Y., Sattar, S., Morris, P., Johnson, K. J., & Burton, H. V. and. (2026). Functional recovery design for new buildings: An integrated economic and engineering evaluation. *Earthquake Spectra, Forthcoming*.

2. Li, Z., Calvo-Bartolomé, L., Hoyle, A. M., Xu, P., Stephens, D. K., Fung, J. F., Dima, A., & Boyd-Graber, J. L. (2025). Large language models struggle to describe the haystack without human help: A social science-inspired evaluation of topic models. In W. Che, J. Nabende, E. Shutova, & M. T. Pilehvar (Eds.), *Proceedings of the 63rd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 7583–7604). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.acl-long.375>
3. Orhan, E., Wein, A. M., Kroll, C. A., & Fung, J. F. (2025). Lessons in business recovery following the 2023 kahramanmara earthquake sequence, türkiye informed by women entrepreneurs. *Earthquake Spectra*, 41(3), 1910–1940. <https://doi.org/10.1177/87552930251330921>
4. Zhang, Y., Fung, J. F., Cook, D., Johnson, K. J., & Sattar, S. (2024). Benefit–cost analysis for earthquake-resilient building design and retrofit: State of the art and future research needs. *Natural Hazards Review*, 25(3), 03124001. <https://doi.org/10.1061/NHREF0.NHENG-1910>
5. Li, Z., Mao, A., Stephens, D., Goel, P., Walpole, E., Dima, A., Fung, J. F., & Boyd-Graber, J. (2024). Improving the TENOR of labeling: Re-evaluating topic models for content analysis. In Y. Graham & M. Purver (Eds.), *Proceedings of the 18th conference of the european chapter of the association for computational linguistics (volume 1: Long papers)* (pp. 840–859). Association for Computational Linguistics. <https://aclanthology.org/2024.eacl-long.51>
6. Zhang, Y., Ayyub, B. M., Fung, J. F., & Labe, Z. M. (2024). Incorporating extreme event attribution into climate change adaptation for civil infrastructure: Methods, benefits, and research needs. *Resilient Cities and Structures*, 3(1), 103–113. <https://doi.org/10.1016/j.rcns.2024.03.002>
7. Fung, J. F., Zhang, Y., Johnson, K. J., Cook, D. T., & Sattar, S. (2023). Multidisciplinary research to advance the development of functional recovery for community resilience. *Disaster Prevention and Resilience*, 2(13). <http://dx.doi.org/10.20517/dpr.2023.15>
8. Zhang, Y., Fung, J. F., Johnson, K. J., & Sattar, S. (2022). Motivators and impediments to seismic retrofit implementation for wood-frame soft-story buildings: A case study in california. *Earthquake Spectra*, 38(4), 2788–2812. <https://doi.org/10.1177/87552930221100844>
9. Thomas, D., & Fung, J. (2022). Measuring downstream supply chain losses due to power disturbances. *Energy Economics*, 114, 106314. <https://doi.org/10.1016/j.eneco.2022.106314>
10. Zhang, Y., Fung, J. F., Johnson, K. J., & Sattar, S. (2022). Review of seismic risk mitigation policies in earthquake-prone countries: Lessons for earthquake resilience in the united states. *Journal of Earthquake Engineering*, 26(12), 6208–6235. <https://doi.org/10.1080/13632469.2021.1911889>
11. Hariri-Ardebili, M. A., Sattar, S., Johnson, K., Clavin, C., Fung, J., & Ceferino, L. (2022). A perspective towards multi-hazard resilient systems: Natural hazards and pandemics. *Sustainability*, 14(8), 4508. <https://doi.org/10.3390/su14084508>

12. Helgeson, J. F., Aminpour, P., Fung, J. F., Henriquez, A. R., Zycherman, A., Butry, D., Nierenberg, C., & Zhang, Y. (2022). Natural hazards compound COVID-19 impacts on small businesses disproportionately for historically underrepresented group operators. *International Journal of Disaster Risk Reduction*, 72, 102845. <https://doi.org/10.1016/j.ijdr.2022.102845>
13. Zhang, Y., Ayyub, B. M., & Fung, J. F. (2022). Projections of corrosion and deterioration of infrastructure in united states coasts under a changing climate. *Resilient Cities and Structures*, 1(1), 98–109. <https://doi.org/https://doi.org/10.1016/j.rcns.2022.04.004>
14. Fung, J. F., Sattar, S., Butry, D. T., & McCabe, S. L. (2021). The total costs of seismic retrofits: State of the art. *Earthquake Spectra*, 0(0), 87552930211009522. <https://doi.org/10.1177/87552930211009522>
15. Helgeson, J. F., Fung, J. F., & Roa-Henriquez, A. R. (2020). Rationally bounded in a storm of complex events: Small businesses facing natural hazard resilience during a pandemic. *Journal of Behavioral Economics for Policy*, 4(S3), 55–65. <https://ideas.repec.org/a/beh/jbepv1/v4y2020is3p55-65.html>
16. Fung, J. F., & Hsu, C.-L. (2021). A cumulative offer process for supply chain networks. *Review of Economic Design*, 25(1). <https://doi.org/10.1007/s10058-020-00238-z>
17. Fung, J. F., Helgeson, J. F., Webb, D. H., O'Fallon, C. M., & Cutler, H. (2020). Does resilience yield dividends? Co-benefits of investing in increased resilience in cedar rapids. *Economic Systems Research*, 1–27. <https://doi.org/10.1080/09535314.2020.1798359>
18. Fung, J. F., Sattar, S., Butry, D. T., & McCabe, S. L. (2020). A predictive modeling approach to estimating seismic retrofit costs. *Earthquake Spectra*, 36(2), 579–598. <https://doi.org/10.1177/8755293019891716>
19. Lindt, J. van de, Peacock, W., Mitrani-Reiser, J., Rosenheim, N., Deniz, D., Dillard, M., Tomiczek, T., Graettinger, A., Crawford, P., Harrison, K., Barbosa, A., Tobin, J., Helgeson, J., Peek, L., Memari, M., Sutley, E., Hamideh, S., Gu, D., Cauffman, S., & Fung, J. (2020). *Community resilience-focused technical investigation of the 2016 lumberton, north carolina flood: An interdisciplinary approach*.
20. Johnson, K. J., Fung, J. F., McAllister, T. P., McCabe, S. L., Sattar, S., & Segura Jr, C. L. (2020). Social and economic components of resilient multihazard building design. *Natural Hazards Review*, 21(1), 6019002.
21. Fung, J. F., Cousins, T., Wein, A., Brown-Giammanco, T., & Helgeson, J. (2026, July). Data collection best practices for post-earthquake reconnaissance: An LLM approach. *Proceedings of the 13th National Conference in Earthquake Engineering*.
22. Wein, A., Orhan, E., Fung, J. F., & Kroll, C. (2026, July). Business and labor perspectives after the 2023 Kahramara, Türkiye earthquakes: Insights on community recovery. *Proceedings of the 13th National Conference in Earthquake Engineering*.
23. Fung, J. F. (2026, July). Building the empirical foundation for functional recovery design. *Proceedings of the 13th National Conference in Earthquake Engineering*.

Publications (other)

1. Thomas, D., & Fung, J. (2023). *Power disturbances: An examination of short-term losses in the downstream supply chain*. AEA Conference, New Orleans, LA, US. https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=935956

2. Fung, J., Sattar, S., Butry, D., & McCabe, S. (2020). *Machine learning methods for predicting seismic retrofit costs*. Proceedings of the 17th World Conference on Earthquake Engineering, Sendai,. https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=929595
3. Fung, J., Helgeson, J., O'Fallon, C., Webb, D., & Cutler, H. (2019). *Quantifying macroeconomic resilience dividends in cedar rapids*. The 27th International Input-Output Association Conference, Glasgow, -1. https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=927870
4. Fung, J., Sattar, S., Butry, D., & McCabe, S. (2019). *Selecting building characteristics to predict seismic retrofit costs of a building portfolio*. Proceedings of the 2nd Annual Conference on Natural Hazards & Infrastructure, Chania, -1. https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=927227
5. Fung, J., Butry, D., Sattar, S., & McCabe, S. (2018). *Cost estimates for the seismic retrofit of federally owned and leased buildings*. Proceedings of the 11th National Conference in Earthquake Engineering, Los Angeles, CA. https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=924676
6. Helgeson, J., Fung, J., O'Fallon, C., Webb, D., & Cutler, H. (2017). *Identifying and quantifying the resilience dividend using computable general equilibrium models: A methodological overview*. Identifying; Quantifying the Resilience Dividend using Computable General Equilibrium Models: A Methodological Overview, Brussels, -1. https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=924427
7. Zhang, Y., Nema, A., Cook, D., Fung, J., & Sattar, S. (2026). *Probabilistic carbon assessment of post-disaster building repairs using pelicun-LCA*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.2374>
8. Gilbert, S., FANG, H., Butry, D., Tam, W. C., Donnelly, M., & Fung, J. (2026). *Understanding the risk of lithium-ion battery fires - multi-source data analysis*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.2365>
9. Fung, J., Stephens, D. K., Dima, A., & Majurski, M. (2026). *Validating LLM-generated data grounded in technical documents: An application in community planning*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.2361>
10. Fung, J., Hall, M., Helgeson, J., Dima, A., Andrews, A., & Benkstein, J. (2025). *Evaluating the impact of changes to the internal publication review policy at NIST*. Data Collection Instruments, National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.DCI.007>
11. Elsibaie, S., Zhang, Y., Fung, J., Cook, D., Sattar, S., Morris, P., Burton, H., Welch, D., & Anaraki, K. (2025). *Estimating the costs and performance of enhanced building design: A case study of functional recovery for new buildings*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.2351>
12. Nikolaou, A., Tsatsis, A., Antoniou, M., Fung, J., Saadat, Y., Gelogoti, F., Kourkoulis, R., & McCabe, S. (2025). Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1340>

13. Fung, J. F., Li, Z., Stephens, D., Mao, A., Goel, P., Walpole, E., Dima, A. A., & Boyd-Graber, J. (2024). *Human-in-the-loop technical document annotation: Developing and validating a system to provide machine-assistance for domain-specific text analysis*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.2287>
14. Zhang, Y., Sattar, S., Cook, D., Johnson, K., & Fung, J. (2024). *Systematic review of embodied carbon assessment and reduction in building life cycles*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1324>
15. Fung, J., Zhang, Y., Johnson, K., Cook, D., & Sattar, S. (2022). *A framework to evaluate the cost-effectiveness of recovery-based design*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1277>
16. Helgeson, J., Fung, J., Henriquez, A. R., Zycherman, A., Mohammadabadi, P. A., Nierenberg, C., Butry, D., & Ramkissoon, D. (2021). *Eliciting lessons from small- and medium-sized enterprises for resilience during and following complex events: Longitudinal data collection (wave 2)*. Data Collection Instruments, National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.DCI.003>
17. Helgeson, J., Fung, J., Henriquez, A. R., Zycherman, A., Nierenberg, C., Butry, D., Ramkissoon, D., & Zhang, Y. (2021). *Longitudinal study of complex event resilience of small- and medium-sized enterprises: Natural disaster planning and recovery during the COVID-19 pandemic (wave 2)*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1267>
18. Helgeson, J., Fung, J., Roa, A., Zycherman, A., Butry, D., Nierenberg, C., Zhang, Y., & Ramkissoon, D. (2020). *Respondent summary report business survey: COVID-19 impacts and recovery in the context of complex events*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1264>
19. Helgeson, J., Fung, J., Zhang, Y., Roa, A., Zycherman, A., Nierenberg, C., Butry, D., & Ramkissoon, D. (2020). *Eliciting lessons from small- and medium-sized enterprises (SMEs) for natural disaster resilience planning and recovery during the COVID-19 pandemic: SME complex event resilience*. Other, National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.DCI.002>
20. Helgeson, J., Fung, J., Roa, A., Zhang, Y., Zycherman, A., Nierenberg, C., Butry, D., & Ramkissoon, D. (2020). *Complex event resilience of small- and medium-sized enterprises: Natural disaster planning during the COVID-19 pandemic*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1258>
21. Helgeson, J., Fung, J., Roa, A., Zhang, Y., Zycherman, A., Nierenberg, C., Butry, D., & Ramkissoon, D. (2020). *Complex event resilience of small- and medium-sized enterprises: Natural disaster planning during the COVID-19 pandemic briefing document*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1259>

22. Gilbert, S., Helgeson, J., Webb, D., Fung, J., & Kandaswamy, A. (2020). *Associating disaster deaths with risk profiles*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.2115>
23. Lindt, J. van de, Peacock, W., Mitrani-Reiser, J., Rosenheim, N., Deniz, D., Dillard, M., Tomiczek, T., Koliou, M., Graettinger, A., Crawford, P., Harrison, K., Barbosa, A., Tobin, J., Helgeson, J., Peek, L., Memari, M., Sutley, E., Hamideh, S., Gu, D., ... Fung, J. (2018). *The lumberton, north carolina flood of 2016: A community resilience focused technical investigation*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1230>
24. Helgeson, J., Fung, J., O'Fallon, C., Webb, D., & Cutler, H. (2018). *A computable general equilibrium model of cedar rapids*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.2029>
25. Sattar, S., McAllister, T., Johnson, K., Clavin, C., Segura, C., McCabe, S., Fung, J., Abrahams, L., Emily, Levitan, M., Harrison, K., & Harris, J. (2018). *Research needs to support immediate occupancy building performance following natural hazard events*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1224>
26. Fung, J., Butry, D., Sattar, S., & McCabe, S. (2018). *Estimating structural seismic retrofit costs for federal buildings*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.1996>
27. Thomas, D., Butry, D., Gilbert, S., Webb, D., & Fung, J. (2017). *The costs and losses of wildfires*. Special Publication (NIST SP), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.SP.1215>
28. Fung, J., Butry, D., Sattar, S., & McCabe, S. (2017). *A methodology for estimating seismic retrofit costs*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.1973>
29. Fung, J., & Helgeson, J. (2017). *Defining the resilience dividend: Accounting for co-benefits of resilience planning*. Technical Note (NIST TN), National Institute of Standards; Technology, Gaithersburg, MD. <https://doi.org/https://doi.org/10.6028/NIST.TN.1959>
30. Roa-Henriquez, A., Fung, J. F., Abud, R., Helgeson, J., & Thomas, D. (2026). Causal machine learning: An empirical approach to supply chain management. In *Digital transformation of the supply chain* (pp. 327–358). Springer. <https://doi.org/https://doi.org/10.1007/978-3-032-01218-0>
31. Butry, D. T., Thomas, D., & Fung, J. F. (2021). Resilience economics and risk management. In B. M. Ayyub (Ed.), *Hazard-resilient infrastructure: Analysis and design*. American Society of Civil Engineers Reston, VA; American Society of Civil Engineers.

Software

NIST Public Data Repository

TOVA: Topic Visualization & Analysis

- doi:10.18434/mds2-4134

python

2026

NIST Public Data Repository <i>RetrofitCostML</i> doi:10.18434/mds2-3971	python 2026
NIST Public Data Repository <i>frbca</i> doi:10.18434/mds2-3966	R 2026
NIST Public Data Repository <i>Understanding the Risk of Lithium Ion Battery Fires</i> doi:10.18434/mds2-4024	R 2026

Leadership, activities, and other skills

Leadership: Earthquake Engineering Research Institute (EERI): Co-chair, Learning From Earthquakes (LFE) Business Resilience Subcommittee.

Professional Organizations: American Economic Association (AEA); American Society of Civil Engineers (ASCE); Earthquake Engineering Research Institute (EERI).

Mentoring: Formal mentoring through NIST Mentoring Program (2023, 2024). Informal mentoring through NIST academic programs (GMSE, SURF, PREP).

Programming: R, Julia, Python, SQL. Instructor for NIST Software Carpentry. Lesson maintainer for Data Carpentry.

Languages: English (fluent), Spanish (native).

Clubs: NIST Toastmasters Club; Association of NIST Hispanic Americans; NIST Community Building Group.

Local Governance: Elected to lead HOA representing neighborhood households; manage meetings, budgets, and vendor oversight.